

Industry Problem Task – AI/ML Healthcare

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO A healthcare company has patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history. It wants a predictive model for early diabetes diagnosis and an AI agent for personalised actions: Diet, Exercise, Medication and Monitor.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING (a) Describe the inductive learning approach. Identify the target and at least three key features with justification. (b) Explain class imbalance and justify SMOTE, under-sampling or class weights. (c) Split 70:30 and justify it. State normalization and encoding steps and their impact.

TASK 2: DECISION TREE FOR DIAGNOSIS (a) Build a Decision Tree and draw its first three levels. Justify the root using Information Gain/Gini. (b) Report Accuracy, Precision, Recall and F1. Identify False Negatives and suggest how to reduce them clinically. (c) Apply cost-complexity post-pruning. Compare pre- and post-pruning and select a deployment model.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION (a) Apply Naive Bayes or Logistic Regression. Compare Accuracy and AUC-ROC with the Decision Tree. Discuss when a statistical model is preferable. (b) Identify top 3 predictors from both models. Do they agree?

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION (a) Define an MDP: states, actions, reward and transition dynamics. (b) Apply Q-Learning for at least 5 patient episodes. Show at least 3 Q-table updates across 2 iterations. State convergence criterion. (c) Extract the final policy and discuss patient safety, bias and explainability.